OS & SP

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SKILL – 14

1.Skill14.c

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/wait.h>

int main(void)
{
   int output_fd;
   int error_fd;
   pid_t pid;

   /* Open append.txt in append mode */
```

```
output_fd = open(
    "append.txt",
    O_WRONLY | O_CREAT | O_APPEND,
    0644
);

if (output_fd == -1) {
    perror("Error opening append.txt");
    return 1;
}

/* Open errors.txt */
error_fd = open(
    "errors.txt",
    O_WRONLY | O_CREAT | O_TRUNC,
    0644
);
```

```
if (error_fd == -1) {
    perror("Error opening errors.txt");
    close(output_fd);
    return 1;
}

/* Create child process */
pid = fork();

if (pid < 0) {
    perror("fork");
    close(output_fd);
    close(error_fd);
    return 1;
}

/* Child process */
if (pid == 0) {
```

```
/* Redirect stdout to append.txt */
if (dup2(output_fd, STDOUT_FILENO) == -1) {
    perror("dup2 stdout");
    _exit(1);
}

/* Redirect stderr to errors.txt */
if (dup2(error_fd, STDERR_FILENO) == -1) {
    perror("dup2 stderr");
    _exit(1);
}

/* Close original descriptors */
close(output_fd);
close(error_fd);

/* Write directly to stdout */
dprintf(
```

```
        STDOUT_FILENO,
        "This line is appended to append.txt\n"
    );

    /* Execute ls on a nonexistent directory */
    execlp(
        "ls",
        "ls",
        "/directory_that_does_not_exist",
        (char *)NULL
    );

    /* Runs only if execlp fails */
    perror("execlp");
    _exit(1);
}

/* Parent process */
```

```
    close(output_fd);
    close(error_fd);

    /* Wait for child */
    waitpid(pid, NULL, 0);

    printf("Q14 completed successfully.\n");
    printf("stdout -> append.txt\n");
    printf("stderr -> errors.txt\n");

    return 0;
}
```

OUTPUT :

```
ramyasree@DESKTOP-DP2O9U6:~/Skill$ nano skill14.c
ramyasree@DESKTOP-DP2O9U6:~/Skill$ gcc -Wall -Wextra -g skill14.c -o skill14
ramyasree@DESKTOP-DP2O9U6:~/Skill$ ./skill14
Q14 completed successfully.
stdout -> append.txt
stderr -> errors.txt
ramyasree@DESKTOP-DP2O9U6:~/Skill$
```